

(*Q*) for all m : the width-sum generating function of a box slice
is PF₂, by per-bead homogenisation and Lorentzian closure

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Abstract

Let $1 \leq L_i \leq U_i$ be integers, $\text{Box} = \prod_{i=1}^m [L_i, U_i] \subset \mathbb{Z}^m$, $T \in \mathbb{Z}$, and $[w]_z = 1 + z + \dots + z^{w-1}$. Statement (*Q*) asserts that $F(z) = \sum_{w \in \text{Box}, \sum_i w_i = T} \prod_i [w_i]_z$ is PF₂, i.e. has log-concave coefficients and no internal zeros. It was known for $m = 2$ by a concavity argument that provably stops at $m = 3$. We prove (*Q*) for every m .

The proof homogenises the generating function *bead by bead* into a polynomial $\prod_{i=1}^m \tilde{P}_i(X, E, W)$ in three variables, all of whose coefficients are 1 and each of whose supports is a box intersected with a hyperplane. Three steps close it. (A) Each $\text{supp } \tilde{P}_i$ is M -convex (Lemma 7), so the constant function $\log c_\alpha \equiv 0$ is M -concave and Brändén–Huh’s corollary **normalizedcoefficients** makes $N(\tilde{P}_i)$ Lorentzian. (B) Brändén–Huh’s corollary **CorollaryConvolution** propagates this along the product. (C) The Hessian of $\partial^\beta N(f)$ has entries the *raw* coefficients $c_{\beta+e_i+e_j}$ — Lemma 14, proved here for all n and all degrees with every factorial displayed — and at $\beta = (a-1, D-a-1, H-D)$ its $\{X, E\}$ minor is $k(a)^2 - k(a-1)k(a+1)$.

Four things worth separating from the headline. First, the factorial cancellation of step (C) is independent of m : m is a count of *factors*, never of variables, and the polynomial lives in three variables for every m . The brief’s hypothesis that the cancellation might hold only for $m \leq 2$ is therefore false, and Remark 15 says why. Second, the route does *not* use the M -convexity theorem for $\Omega \subset \mathbb{Z}^{2m}$, whose single hypothesis was laminarity of the pair-constraint family. Route 2 needs only the box-slice case, Lemma 7, whose proof consumes nothing but coordinatewise bounds — and in which *every* admissible index is an exchange partner. The dependency set is strictly smaller than the brief assumed (Remark 10). Third, a number in the brief that set this session was bound to the wrong theorem, and Remark 21 separates them: (*Q*) closes condition (A) on the 12,665 *one-sided* slices, not on all 20,322 single-half-width slices, because on the other 7657 the bridge — not (*Q*) — is what is missing. Fourth, the proof rests on one line of an e-print that could not be reopened in this session. Section 7 states that dependency exactly, separates the two Brändén–Huh reads that support it, and reports a refusal control — 779,641 pairs, 211,598 of them outside the reach of the other cited corollary, zero counterexamples — aimed at the recorded *statement* rather than at (*Q*).

1 Statement and conventions

Definition 1 (PF₂). A finitely supported sequence $(c_a)_{a \in \mathbb{Z}}$ of nonnegative reals is PF₂ if its support is an interval of integers and $c_a^2 \geq c_{a-1}c_{a+1}$ for all a . A polynomial is PF₂ if its coefficient sequence is.

Throughout, $m \geq 1$, and $1 \leq L_i \leq U_i$ are integers for $1 \leq i \leq m$, $\text{Box} = \prod_i [L_i, U_i] \subset \mathbb{Z}^m$, and $T \in \mathbb{Z}$. Write

$$W_T = \{w \in \text{Box} : \sum_i w_i = T\}, \quad F(z) = \sum_{w \in W_T} \prod_{i=1}^m [w_i]_z, \quad [w]_z = 1 + z + \dots + z^{w-1}.$$

It is convenient to shift: put

$$l_i = L_i - 1, \quad h_i = U_i - 1, \quad D = T - m, \quad H = \sum_{i=1}^m h_i,$$

so $0 \leq l_i \leq h_i$, and $W_T \neq \emptyset$ if and only if $\sum_i l_i \leq D \leq H$.

Proposition 2 (lattice-point form). *For every $a \in \mathbb{Z}$,*

$$[z^a]F(z) = k(a) := \#\left\{(x, e) \in \mathbb{Z}_{\geq 0}^m \times \mathbb{Z}_{\geq 0}^m : l_i \leq x_i + e_i \leq h_i \ (1 \leq i \leq m), \sum_i x_i = a, \sum_i e_i = D - a\right\}.$$

Proof. Expanding each $[w_i]_z = \sum_{x_i=0}^{w_i-1} z^{x_i}$,

$$[z^a]F(z) = \#\{(w, x) : w \in \text{Box}, \sum_i w_i = T, 0 \leq x_i \leq w_i - 1, \sum_i x_i = a\}.$$

Put $u_i = w_i - 1$ and $e_i = u_i - x_i$. Then $u \in \prod_i [l_i, h_i]$, $\sum_i u_i = T - m = D$, $x_i, e_i \geq 0$, $x_i + e_i = u_i$, and $\sum_i e_i = D - a$; the map $(w, x) \mapsto (x, e)$ is a bijection onto the set counted by $k(a)$, with inverse $w_i = x_i + e_i + 1$. $\square$

Proposition 3 (interval support, for free). *If $W_T \neq \emptyset$ then $\text{supp } F = [0, D]$ exactly. Consequently (Q) is equivalent to*

$$k(a)^2 \geq k(a-1)k(a+1) \quad \text{for all } 1 \leq a \leq D-1. \quad (Q')$$

Proof. For a single $w \in W_T$ the polynomial $\prod_i [w_i]_z$ has support exactly $[0, \sum_i (w_i - 1)] = [0, D]$, because each factor has support $[0, w_i - 1]$ with $w_i \geq L_i \geq 1$. A sum of nonnegative sequences with common support $[0, D]$ has support $[0, D]$. Since the support is then a full interval, the no-internal-zeros half of PF_2 holds, and only the log-concavity inequalities remain. $\square$

This is the point at which the two halves of PF_2 part company, and it is worth saying once, loudly: PF_2 is log-concavity **and** interval support, and the two halves are proved here by two unrelated arguments. Interval support comes from Proposition 3, which looks at one summand at a time and never mentions Lorentzian polynomials. Log-concavity comes from Theorem 19 below, which never looks at a single summand. A Hessian minor delivers log-concavity and nothing else; on its own it does not give PF_2 .

Discrete convexity and Lorentzian polynomials

Definition 4 (M -convex, M -concave). A nonempty finite $S \subset \mathbb{Z}^n$ is M -convex if it satisfies the symmetric exchange axiom: for all $\alpha, \beta \in S$ and every i with $\alpha_i > \beta_i$ there is a j with $\alpha_j < \beta_j$ such that $\alpha - e_i + e_j \in S$ and $\beta + e_i - e_j \in S$. A function $\nu : \mathbb{Z}^n \rightarrow \mathbb{R} \cup \{-\infty\}$ with finite nonempty $\text{dom } \nu = \{\alpha : \nu(\alpha) > -\infty\}$ is M -concave if for all $\alpha, \beta \in \text{dom } \nu$ and every i with $\alpha_i > \beta_i$ there is a j with $\alpha_j < \beta_j$ such that

$$\nu(\alpha) + \nu(\beta) \leq \nu(\alpha - e_i + e_j) + \nu(\beta + e_i - e_j).$$

Taking ν to be the $0/-\infty$ indicator of S recovers M -convexity of S from M -concavity of ν .

Lemma 5 (M -convex sets have constant coordinate sum). *If $S \subset \mathbb{Z}^n$ is M -convex then $\sum_i \alpha_i$ is the same for all $\alpha \in S$.*

Proof. Suppose $\alpha, \beta \in S$ with $\sum_i \alpha_i > \sum_i \beta_i$. Put $d(\alpha, \beta) = \sum_i (\alpha_i - \beta_i)_+$. Since $\sum \alpha > \sum \beta$ there is an i with $\alpha_i > \beta_i$, so $d(\alpha, \beta) \geq 1$; the exchange axiom supplies j with $\alpha_j < \beta_j$ and $\alpha' = \alpha - e_i + e_j \in S$. Then $(\alpha'_i - \beta_i)_+ = (\alpha_i - \beta_i)_+ - 1$ and $(\alpha'_j - \beta_j)_+ = 0 = (\alpha_j - \beta_j)_+$ because $\alpha_j + 1 \leq \beta_j$, while all other coordinates are unchanged; so $d(\alpha', \beta) = d(\alpha, \beta) - 1$, and $\sum_i \alpha'_i = \sum_i \alpha_i$. Iterating, we reach some $\alpha'' \in S$ with $d(\alpha'', \beta) = 0$, i.e. $\alpha'' \leq \beta$, whence $\sum_i \alpha''_i \leq \sum_i \beta_i < \sum_i \alpha_i = \sum_i \alpha''_i$, a contradiction. $\square$

Lemma 5 is the reason the route must *homogenise*, and it is worth saying before the construction rather than after. The set $\{(x, e) \in \mathbb{Z}_{\geq 0}^2 : l \leq x + e \leq h\}$ that the problem hands us does *not* have constant coordinate sum when $l < h$, so it cannot be M -convex and no amount of care will make `normalizedcoefficients` apply to it. Introducing the slack coordinate $w = h - x - e$ is exactly what buys the constant sum. The third variable W is not bookkeeping; it is the hypothesis.

Write H_n^d for the homogeneous polynomials of degree d in $v_1, \dots, v_n$ with nonnegative coefficients. For $f = \sum_{\alpha} c_{\alpha} v^{\alpha} \in H_n^d$ put

$$N(f) = \sum_{\alpha} \frac{c_{\alpha}}{\alpha!} v^{\alpha}, \quad \alpha! = \prod_i \alpha_i!.$$

We use Brändén–Huh’s class L_n^d of *Lorentzian* polynomials in the form recorded in §7: L_n^d is the closure of the strictly Lorentzian polynomials, which are defined by: $f \in H_n^d$ with all coefficients positive is strictly Lorentzian iff $\text{Hess } f$ is nonsingular with exactly one positive eigenvalue, and for $d > 2$, f is strictly Lorentzian iff all $\partial_i f$ are.

Lemma 6 (what we use of the definition). *If $f \in L_n^d$ and $\beta \in \mathbb{Z}_{\geq 0}^n$ with $|\beta| = d - 2$, then the Hessian of $\partial^{\beta} f$ has at most one positive eigenvalue.*

Proof. By the recursive clause, ∂^{β} maps strictly Lorentzian polynomials of degree d to strictly Lorentzian polynomials of degree 2, whose Hessians have exactly one positive eigenvalue. The map $f \mapsto \text{Hess}(\partial^{\beta} f)$ is linear, hence continuous, and the condition “ $\lambda_2 \leq 0$ ” on a symmetric matrix is closed (the eigenvalues depend continuously on the matrix). So the condition passes to the closure L_n^d . $\square$

2 Step A: the beads

Lemma 7 (box slices are M -convex, with every index an exchange partner). *Let $p, q \in \mathbb{Z}^n$ with $p \leq q$, let $d \in \mathbb{Z}$, and let*

$$B = \{\alpha \in \mathbb{Z}^n : p \leq \alpha \leq q, \sum_i \alpha_i = d\}.$$

If $B \neq \emptyset$ then B is M -convex. Moreover, for all $\alpha, \beta \in B$ and every i with $\alpha_i > \beta_i$, every j with $\alpha_j < \beta_j$ is an admissible exchange partner, and at least one such j exists.

Proof. Let $\alpha, \beta \in B$ and let i satisfy $\alpha_i > \beta_i$. Since $\sum_j \alpha_j = \sum_j \beta_j = d$ and $\alpha_i - \beta_i > 0$, the set $\{j : \alpha_j < \beta_j\}$ is nonempty. Fix any such j (necessarily $j \neq i$). Then

$$\alpha_i - 1 \geq p_i \quad (\text{because } \alpha_i > \beta_i \geq p_i), \quad \alpha_j + 1 \leq q_j \quad (\text{because } \alpha_j < \beta_j \leq q_j),$$

so $\alpha - e_i + e_j$ lies in the box; its coordinate sum is still d ; hence $\alpha - e_i + e_j \in B$. Symmetrically

$$\beta_i + 1 \leq q_i \quad (\text{because } \beta_i < \alpha_i \leq q_i), \quad \beta_j - 1 \geq p_j \quad (\text{because } \beta_j > \alpha_j \geq p_j),$$

so $\beta + e_i - e_j \in B$. Each of the four inequalities is obtained from one strict inequality between α and β together with one box constraint, and each is used exactly once. $\square$

Definition 8 (the bead polynomials). For $0 \leq l \leq h$ put

$$\tilde{P}^{(l,h)}(X, E, W) = \sum_{\substack{x,e,w \geq 0, \ x+e+w=h \\ l \leq x+e \leq h}} X^x E^e W^w,$$

and write $\tilde{P}_i = \tilde{P}^{(l_i, h_i)}$.

Lemma 9 (Step A). *Let $0 \leq l \leq h$ and $f = \tilde{P}^{(l,h)}$. Then*

- (a) *f is homogeneous of degree h in three variables and all its coefficients equal 1;*
- (b) *$\text{supp } f = \{(x, e, w) \in \mathbb{Z}_{\geq 0}^3 : x + e + w = h, w \leq h - l\}$, which is the box slice of Lemma 7 with $n = 3$, $p = (0, 0, 0)$, $q = (h, h, h - l)$, $d = h$; in particular $\text{supp } f$ is M -convex;*
- (c) *$\nu_f(\alpha) = \log c_\alpha$ equals 0 on $\text{supp } f$ and $-\infty$ off it, and is M -concave;*
- (d) *$N(f)$ is Lorentzian.*

Proof. (a) Every monomial in the defining sum has total degree $x + e + w = h$, and distinct index triples give distinct monomials, so each coefficient is 0 or 1.

(b) Under $x + e + w = h$ with $x, e, w \geq 0$ the condition $x + e \leq h$ is automatic ($x + e = h - w \leq h$), and $x + e \geq l$ says $h - w \geq l$, i.e. $w \leq h - l$. The bounds $x \leq h$ and $e \leq h$ are likewise automatic. So the support is as claimed, and it is exactly $\{\alpha \in \mathbb{Z}^3 : 0 \leq \alpha \leq (h, h, h - l), \alpha_1 + \alpha_2 + \alpha_3 = h\}$. It is nonempty (it contains $(h, 0, 0)$), so Lemma 7 applies.

(c) All coefficients on the support are 1, so $\nu_f \equiv 0$ there. Given $\alpha, \beta \in \text{supp } f$ and i with $\alpha_i > \beta_i$, Lemma 7 supplies a j with $\alpha_j < \beta_j$ and both exchanged points in $\text{supp } f$; for those points

$$\nu_f(\alpha) + \nu_f(\beta) = 0 = \nu_f(\alpha - e_i + e_j) + \nu_f(\beta + e_i - e_j).$$

So the M -concavity inequality holds, with equality.

(d) By (c) and Brändén–Huh **normalized coefficients** (§7), the polynomial $\sum_\alpha (c_\alpha / \alpha!) v^\alpha = N(f)$ is Lorentzian. $\square$

Remark 10 (where laminarity went). The programme’s previous M -convexity theorem concerned $\Omega = \{(x, e) \in \mathbb{Z}_{\geq 0}^{2m} : l_i \leq x_i + e_i \leq h_i, \sum x + \sum e = D\} \subset \mathbb{Z}^{2m}$, and its proof consumed exactly one hypothesis: laminarity of the constraint family (singletons, the pairs $\{x_i, e_i\}$, the ground set).

Route 2 does not use that theorem at all. The only M -convexity it needs is Lemma 7 applied in $\mathbb{Z}^3$ to a plain box slice, and the proof of Lemma 7 mentions no constraint family: it uses coordinatewise bounds and nothing else, and it finds that *every* index with $\alpha_j < \beta_j$ works, where the pair-constrained proof had to choose j by a case split on the sign of $v_1 - v'_1$. So the honest statement is not “laminarity is unused” — a box’s constraint family is trivially laminar — but the stronger one: *the theorem whose hypothesis was laminarity is not a premise of this proof*, and the hypothesis that replaces it is weaker and m -free. This is why the route works for every m while the $m = 2$ concavity route does not.

3 The product form

Proposition 11 (product form). *Assume $\sum_i l_i \leq D \leq H$ and set $f = \prod_{i=1}^m \tilde{P}_i$, a homogeneous polynomial of degree H in (X, E, W) with nonnegative integer coefficients c_α . Then for every a with $0 \leq a \leq D$,*

$$k(a) = c_{(a, D-a, H-D)} = [X^a E^{D-a} W^{H-D}] \prod_{i=1}^m \tilde{P}_i.$$

Proof. Since all bead coefficients are 1 (Lemma 9(a)),

$$c_{(a,b,j)} = \#\left\{(x, e, w) \in \mathbb{Z}_{\geq 0}^{3m} : x_i + e_i + w_i = h_i, w_i \leq h_i - l_i \forall i, \sum_i x_i = a, \sum_i e_i = b, \sum_i w_i = j\right\}.$$

Put $b = D - a$ and $j = H - D$. For a triple satisfying the per-bead equations, $\sum_i (x_i + e_i) + \sum_i w_i = H$, so the condition $\sum_i w_i = H - D$ is equivalent to $\sum_i (x_i + e_i) = D$, which given $\sum x_i = a$ is equivalent to $\sum e_i = D - a$. Hence w is determined by (x, e) via $w_i = h_i - x_i - e_i$, the constraint $0 \leq w_i \leq h_i - l_i$ is equivalent to $l_i \leq x_i + e_i \leq h_i$, and the count becomes

$$\#\{(x, e) \in \mathbb{Z}_{\geq 0}^{2m} : l_i \leq x_i + e_i \leq h_i, \sum_i x_i = a, \sum_i e_i = D - a\} = k(a). \quad \square$$

Remark 12 (beads of degree zero). If $h_i = 0$ then $l_i = 0$ and $\tilde{P}_i = 1$; such a bead contributes neither to F nor to $k(\cdot)$ (it forces $x_i = e_i = 0$, i.e. $w_i = 1$, $[w_i]_z = 1$) and may be deleted, together with the corresponding coordinate, without changing anything. So we may and do assume $h_i \geq 1$ for all i , unless $m = 0$, in which case $D = 0$, $F = 1$ and (Q) is trivial.

4 Step B: along the product

Proposition 13 (Step B). $N(\prod_{i=1}^m \tilde{P}_i)$ is Lorentzian.

Proof. Induction on m . For $m = 1$ this is Lemma 9(d). Let $m \geq 2$, put $f' = \prod_{i < m} \tilde{P}_i$ and $g = \tilde{P}_m$. Both are homogeneous with nonnegative integer coefficients in the same three variables X, E, W , of degrees $\sum_{i < m} h_i$ and h_m respectively. By the inductive hypothesis $N(f')$ is Lorentzian, and by Lemma 9(d) so is $N(g)$. By Brändén–Huh CorollaryConvolution (§7), $N(f'g) = N(\prod_{i \leq m} \tilde{P}_i)$ is Lorentzian. $\square$

Two hypotheses of the cited corollary are checked explicitly above: homogeneity with nonnegative coefficients, and a shared variable set (all $\tilde{P}_i$ are polynomials in the same X, E, W — this is the whole point of homogenising per bead rather than per coordinate). A third is checked by the shape of the induction: only the *two-factor* form of the corollary is invoked, so it does not matter whether the source states it for two factors or for finitely many. What is *not* checked, and cannot be in this session, is the verbatim hypothesis list at the cited line; see §7.

5 Step C: the factorials

Lemma 14 (raw-Hessian lemma). Let $n \geq 1$, $d \geq 2$, and $f = \sum_{|\alpha|=d} c_\alpha v^\alpha \in H_n^d$. Let $\beta \in \mathbb{Z}_{\geq 0}^n$ with $|\beta| = d - 2$. Then

$$\partial^\beta N(f) = \frac{1}{2} v^\top M(\beta) v, \quad M(\beta)_{ij} = c_{\beta+e_i+e_j},$$

and consequently $\text{Hess}(\partial^\beta N(f)) = M(\beta)$: the entries of the Hessian are the raw coefficients of f , with no residual factorial.

Proof. We display every factorial. By definition $N(f) = \sum_{|\alpha|=d} \frac{c_\alpha}{\alpha!} v^\alpha$, and for a single monomial

$$\partial^\beta v^\alpha = \begin{cases} \frac{\alpha!}{(\alpha - \beta)!} v^{\alpha - \beta}, & \alpha \geq \beta, \\ 0, & \text{otherwise.} \end{cases}$$

Hence

$$\partial^\beta N(f) = \sum_{\substack{|\alpha|=d \\ \alpha \geq \beta}} \frac{c_\alpha}{\alpha!} \cdot \frac{\alpha!}{(\alpha - \beta)!} v^{\alpha - \beta} = \sum_{\substack{|\alpha|=d \\ \alpha \geq \beta}} \frac{c_\alpha}{(\alpha - \beta)!} v^{\alpha - \beta}.$$

The $\alpha!$ introduced by N and the $\alpha!$ produced by ∂^β cancel exactly; what survives is $1/(\alpha - \beta)!$. Substituting $\gamma = \alpha - \beta$, which satisfies $|\gamma| = d - (d - 2) = 2$ and $\gamma \geq 0$,

$$\partial^\beta N(f) = \sum_{|\gamma|=2} \frac{c_{\beta+\gamma}}{\gamma!} v^\gamma.$$

The vectors $\gamma \geq 0$ with $|\gamma| = 2$ are $\gamma = 2e_i$, for which $\gamma! = 2! = 2$ and $v^\gamma = v_i^2$, and $\gamma = e_i + e_j$ with $i < j$, for which $\gamma! = 1! \cdot 1! = 1$ and $v^\gamma = v_i v_j$. Therefore

$$\partial^\beta N(f) = \sum_i \frac{c_{\beta+2e_i}}{2} v_i^2 + \sum_{i < j} c_{\beta+e_i+e_j} v_i v_j = \frac{1}{2} \left(\sum_i M_{ii} v_i^2 + 2 \sum_{i < j} M_{ij} v_i v_j \right) = \frac{1}{2} v^\top M v$$

with $M_{ij} = c_{\beta+e_i+e_j}$, which is symmetric because $e_i + e_j = e_j + e_i$. Finally $\text{Hess}(\frac{1}{2} v^\top M v) = M$ for symmetric M : the $v_i v_j$ -coefficient of $\frac{1}{2} v^\top M v$ is M_{ij} for $i \neq j$ and $\frac{1}{2} M_{ii}$ for $i = j$, and $\partial_i \partial_j$ applied to these returns M_{ij} in both cases. $\square$

Remark 15 (the cancellation is independent of m). The brief that set this session asked for the factorial bookkeeping of step (C) “written out in full, with every factorial visible, for general m ”, and warned that if it only cancels for $m \leq 2$ then the real obstruction has been found. It cancels for every m , and the reason is structural rather than computational: m *never enters Lemma 14*. The parameter m counts the *factors* of $f = \prod_i \tilde{P}_i$; the lemma is a statement about a single homogeneous polynomial, and the only parameters it sees are the number of variables n — which is 3 for every m , because the beads are homogenised into *shared* variables X, E, W — and the degree $d = H$. So there is no m -dependent quantity in the cancellation, and in particular no m at which it could fail.

The contrast with the wrong version of the route is now visible as an ordering of operations. If one applies **normalizedcoefficients** to the indicator of $\Omega \subset \mathbb{Z}^{2m}$ and *then* substitutes X for every x -variable and E for every e -variable, one is evaluating

$$N(\mathbf{1}_\Omega)(X, \dots, X, E, \dots, E) = \sum_{(x,e) \in \Omega} \frac{1}{x! e!} X^{\sum x} E^{\sum e},$$

whose $X^a E^{D-a}$ coefficient is $\sum_{z \in K_a} 1/z!$ and not $|K_a|$. No differentiation has taken place, so nothing has reintroduced the factorials that N divided out. In the per-bead version the factorials are divided out by N and then put back by ∂^β , as the proof of Lemma 14 shows line by line; the substitution that destroys the count is never performed, because the variables were shared from the start.

Proposition 16 (the doubly-corroborated labels cannot do it). *Set $\psi := \prod_{i=1}^m N(\tilde{P}_i)(X, E, W)$. Then ψ is Lorentzian, by **CorollaryProduct** applied in the three shared variables — equivalently, and this is the form in which the temptation presents itself, by embedding the beads in disjoint blocks (X_i, E_i, W_i) , where N does commute with multiplication because $\alpha!$ factorises, and then collapsing the blocks by the nonnegative substitution $X_i \mapsto X$, $E_i \mapsto E$, $W_i \mapsto W$ using **flow**. Either way: no appeal to **CorollaryConvolution** is needed to see that ψ is Lorentzian. Nevertheless:*

(a) $\psi \in \mathbb{Q}[W, X + E]$; explicitly $\psi = \sum_{j \geq 0} \theta_j W^j (X + E)^{H-j}$ for some $\theta_j \geq 0$.

(b) Writing $k_w(\gamma) = \gamma! \cdot [v^\gamma] \psi$ for the raw coefficients to which Lemma 14 applies, $k_w(a, D - a, H - D) = (H - D)! \theta_{H-D}$ is **independent of a** .

So this route proves only that a constant sequence is log-concave. It does not perturb $k(a)$; it annihilates all dependence on a .

Proof. (a) Fix i and write $g_i = h_i - l_i$. By Lemma 9(b) the support of $\tilde{P}_i$ is $\{(x, e, w) : x + e + w = h_i, w \leq g_i\}$ with all coefficients 1, so

$$N(\tilde{P}_i) = \sum_{w=0}^{g_i} \frac{W^w}{w!} \sum_{x+e=h_i-w} \frac{X^x E^e}{x! e!} = \sum_{w=0}^{g_i} \frac{W^w}{w!} \cdot \frac{(X + E)^{h_i-w}}{(h_i - w)!},$$

the inner sum being the binomial theorem. Hence $N(\tilde{P}_i) \in \mathbb{Q}[W, X + E]$, and so is the product ψ . Since ψ is homogeneous of degree H , it is a nonnegative combination $\sum_j \theta_j W^j (X + E)^{H-j}$.

(b) Let $\gamma = (a, b, j)$ with $a + b + j = H$. By (a), $[X^a E^b W^j] \psi = \theta_j \binom{a+b}{a}$, so

$$k_w(\gamma) = a! b! j! \cdot \theta_j \frac{(a+b)!}{a! b!} = j! (a+b)! \theta_j,$$

which depends on γ only through j and $a + b = H - j$. With $j = H - D$ this is $(H - D)! \theta_{H-D}$ for every a . $\square$

Remark 17 (what Proposition 16 settles). This is the precise sense in which the proof's dependence on **CorollaryConvolution** is not an artefact of a careless choice of route. The three closure results that are better corroborated in my records — **CorollaryProduct** (read by both passes at the same line), **flow** and **derivatives** — all act linearly on *normalised* coefficients, and composing them in the one way that reaches three shared variables produces the binomially weighted convolution rather than the count. The weights $\gamma! / \prod_i \alpha^i!$ do not merely distort k : by Proposition 16(b) they flatten it to a constant, because ψ cannot see the X/E split at all — it is a polynomial in $X + E$. The record's own one-line warning, “ N does not commute with multiplication, which is exactly what makes the corollary non-trivial”, is therefore an understatement: the non-commutation carries the entire content of (Q) .

Numerically, with $l = (0, 0)$, $h = (3, 3)$, $D = 4$: the theorem's sequence is $k = (3, 6, 7, 6, 3)$ while $k_w = (20, 20, 20, 20, 20) = \binom{6}{3}$ throughout; with $l = (1, 1, 1)$, $h = (2, 2, 2)$, $D = 4$: $k = (3, 9, 12, 9, 3)$ against $k_w = (72, 72, 72, 72, 72)$.

Lemma 18 (2×2 minors). *Let M be a symmetric real $n \times n$ matrix with at most one positive eigenvalue and with $M_{ii} \geq 0$ for all i . Then $M_{ii} M_{jj} - M_{ij}^2 \leq 0$ for all i, j .*

Proof. For $i = j$ the quantity is $M_{ii} M_{ii} - M_{ii}^2 = 0$, so assume $i \neq j$ and let $A = \begin{pmatrix} M_{ii} & M_{ij} \\ M_{ij} & M_{jj} \end{pmatrix}$, the principal submatrix of M on $\{i, j\}$. Suppose for contradiction $\det A > 0$. Then $M_{ii} M_{jj} > M_{ij}^2 \geq 0$, so M_{ii} and M_{jj} are both nonzero, hence both strictly positive by hypothesis; with $\det A > 0$ this makes A positive definite. Thus $u^\top M u > 0$ for every nonzero u in the 2-dimensional subspace $V = \text{span}\{e_i, e_j\}$.

On the other hand, let $\lambda_1 \geq \dots \geq \lambda_n$ be the eigenvalues of M with an orthonormal eigenbasis $u_1, \dots, u_n$. By hypothesis $\lambda_2 \leq 0$, so $U = \text{span}\{u_2, \dots, u_n\}$ has dimension $n - 1$ and $u^\top M u \leq 0$ for all $u \in U$. Since $\dim V + \dim U = 2 + (n - 1) > n$, there is a nonzero $u \in V \cap U$, and then $u^\top M u > 0$ and $u^\top M u \leq 0$ — a contradiction. Hence $\det A \leq 0$. $\square$

6 The theorem

Theorem 19 ((Q), for all m). *Let $m \geq 1$, let $1 \leq L_i \leq U_i$ be integers, $\text{Box} = \prod_i [L_i, U_i]$, $T \in \mathbb{Z}$, and suppose $W_T = \{w \in \text{Box} : \sum_i w_i = T\} \neq \emptyset$. Then*

$$F(z) = \sum_{w \in W_T} \prod_{i=1}^m [w_i]_z$$

is PF_2 : $\text{supp } F = [0, D]$ with $D = T - m$, and $k(a)^2 \geq k(a-1)k(a+1)$ for all a .

Proof. By Remark 12 we may assume $h_i = U_i - 1 \geq 1$ for all i , so $H = \sum_i h_i \geq m \geq 1$. By Proposition 2, $[z^a]F = k(a)$, and by Proposition 3, $\text{supp } F = [0, D]$; it remains to prove (Q'), i.e. $k(a)^2 \geq k(a-1)k(a+1)$ for $1 \leq a \leq D-1$. If $D \leq 1$ there is no such a and we are done, so assume $D \geq 2$; then $H \geq D \geq 2$.

Put $f = \prod_{i=1}^m \tilde{P}_i$, homogeneous of degree H in (X, E, W) with nonnegative integer coefficients c_α . By Proposition 13, $N(f) \in \mathbb{L}_3^H$.

Fix a with $1 \leq a \leq D-1$ and set

$$\beta = (a-1, D-a-1, H-D) \in \mathbb{Z}_{\geq 0}^3,$$

which is indeed nonnegative ($a \geq 1$; $a \leq D-1$; $D \leq H$) and satisfies $|\beta| = (a-1) + (D-a-1) + (H-D) = H-2$. By Lemma 6 the Hessian of $\partial^\beta N(f)$ has at most one positive eigenvalue, and by Lemma 14 that Hessian is $M = M(\beta)$ with $M_{ij} = c_{\beta+e_i+e_j}$. Index the variables as $1 = X$, $2 = E$, $3 = W$ and read off three entries using Proposition 11:

$$M_{XX} = c_{\beta+2e_X} = c_{(a+1, D-a-1, H-D)} = k(a+1),$$

$$M_{EE} = c_{\beta+2e_E} = c_{(a-1, D-a+1, H-D)} = k(a-1),$$

$$M_{XE} = c_{\beta+e_X+e_E} = c_{(a, D-a, H-D)} = k(a).$$

(The three monomials have the same W -exponent $H-D$; this is exactly why the relevant minor is the one on $\{X, E\}$.) All diagonal entries of M are cardinalities, hence ≥ 0 . Lemma 18 applied to M with $i = X$, $j = E$ gives

$$k(a+1)k(a-1) - k(a)^2 = M_{XX}M_{EE} - M_{XE}^2 \leq 0,$$

which is (Q'). Together with Proposition 3, F is PF_2 . $\square$

Corollary 20 (condition (A) on one-sided slices). *Condition (A) holds on every effective slice that is one-sided, i.e. on which $y_i \geq 0$ for all y and all i , or $y_i \leq 0$ for all y and all i . In the measured range (25,768 nonempty effective slices, $2 \leq m \leq 5$, $m \leq n \leq m+4$, $d \leq 7$) that is 12,665 slices, of which 2,132 are nontrivial.*

Proof. **A-constant-defect-affine-width** (= **thm:affine** of the 1004c2 paper; proved) shows that under one-sidedness the width map $y \mapsto w$ is affine, the shift $\sum_i (-y_i)_+$ is *constant* on the slice, $W = \{w : L_i \leq w_i \leq U_i, \sum_i w_i = T\}$ is a box slice, and $k(a + \text{shift}, b) = \sum_{w \in W} \text{Trap}_w(a)$ is the coefficient sequence of F . So (A) on such a slice is (Q) for that box and T . Apply Theorem 19. $\square$

Remark 21 (a number in my own brief was bound to the wrong theorem). The brief that set this session says that `thm:affine` “reduces condition (A) to exactly (Q) on every *single-half-width* slice — 20,322 of them”, and instructed me to record that closing (Q) closes (A) there. That binding is wrong, and both numbers in it are true of their own object. `thm:affine`’s hypothesis is *one-sidedness*, which holds on 12,665 slices ($y \geq 0$ on 9203, $y \leq 0$ on 3909). The figure 20,322 counts the slices with a single half-width, which are *exactly* the slices where W is a box slice — that is, where (Q)’s own hypothesis holds. The 1004 c2 paper states the relation correctly and in terms (`prop:reach`): “the hypothesis of `thm:affine` is strictly stronger than (Q)’s: 7657 slices have W a box slice without being one-sided”. On those 7657 slices (Q) is available but the *bridge* is not, because without one-sidedness the shift $\sum_i (-y_i)_+$ need not be constant on the slice and $\sum_{w \in W} \text{Trap}_w(a - \text{shift}(w))$ is then not a coefficient of a single F . So Corollary 20 claims 12,665, not 20,322.

Recorded rather than silently corrected, because the defect has the shape my own records keep naming: a figure transplanted between two objects, each proposition true on its own side, nothing in any instrument grading the *binding* — and the artifact that carried it was the brief I wrote to prevent exactly this. Closing the extra 7657 slices needs a constant-shift argument, not more work on (Q); that is a well-posed next target.

We state explicitly what Corollary 20 does *not* do. The parent conjecture `conj-A-logconcave` is untouched on the multi-half-width slices: there the width-vector set is not a box slice, (Q) does not apply, and `A-shape-blind-impossible` already shows that any proof there must consume the full width vectors. A partial result does not relabel the parent.

7 Audit of the chain, and the one dependency that remains

The two Brändén–Huh reads, separated

The proof cites *Lorentzian polynomials*, Brändén–Huh, [arXiv:1902.03719](#), by internal $\text{\LaTeX}$ `\label` and source line number of the e-print `.tex` (4977 lines), read at first hand; provenance is recorded in `memory/reading/sources.json:locators`. Two distinct reads contribute, and it matters which:

label	line	used for	read
<code>FirstDefinition</code>	433	definition of L_n^d (Lemma 6)	2026-09-23 c2
<code>chars</code>	1623	closure of strict $= L_n^d$ (Lemma 6)	2026-09-23 c2
<code>SecondDefinition</code>	637	cross-check of the definition	2026-09-23 c2
<code>CorollaryProduct</code>	1758	<i>not used</i>	both passes
<code>normalizedcoefficients</code>	2853	Lemma 9(d)	2026-09-30 c1
<code>CorollaryConvolution</code>	2416	Proposition 13	2026-09-30 c1

The definitional half of the proof rests on the earlier, long-settled read. The two closure corollaries, which are the load-bearing citations, rest on a single read, recorded in two artifacts (registry node `lorentzian-closure-theorems` prose, preserved in two backups, and `proofs/2026-09-30-c1-cylindric-k` ll.495–528) and merged into `sources.json:locators` on 2026-10-04. Worth naming rather than hiding: the one label corroborated by *two* independent reads at the same line, `CorollaryProduct`, is the one this proof does *not* need. The shared anchor corroborates that the two passes read the same e-print; it does not corroborate the two labels the proof stands on.

What is recorded is each corollary’s *statement*. What is not recorded, and could not be recovered in this no-browsing session, is the verbatim hypothesis list at l.2416. The hypotheses

I can and do check — homogeneity, nonnegative coefficients, a shared variable set, and the two-factor form — are checked in §4. Three further things can be said without reopening the source.

First, the recorded form is internally pinned to the same-variables reading, and this can be settled by reasoning rather than by re-reading. If `CorollaryConvolution` were a statement about f, g in *disjoint* variable blocks, then $N(fg) = N(f)N(g)$ identically — because $\alpha!$ factorises across disjoint blocks — and the statement would be precisely `CorollaryProduct`. The record says in terms that it is *not* an instance of `CorollaryProduct`. So the record is coherent only under the same-variables reading, which is the reading Proposition 13 uses. (Verified: $N(fg) = N(f)N(g)$ on disjoint variables, 54 monomials, exact rational arithmetic; and $N(fg) \neq N(f)N(g)$ on shared variables, 20 monomials.)

Second, the residual risk is precise. If 1.2416 carries a hypothesis strictly stronger than “ $N(f), N(g)$ Lorentzian” — the natural candidate being “ ν_f, ν_g M -concave”, since that is the hypothesis of the neighbouring corollary — then Proposition 13 does not go through as written, because ν of a *product* of beads is not the constant function. Nothing else in the chain is affected: Lemmas 7, 9, 14, 18, Propositions 2, 3, 11 and Theorem 19’s own argument are unconditional.

Third, that worst case is measured, though not proved. Under the stronger reading the induction needs ν of each partial product $\prod_{i \leq k} \tilde{P}_i$ to be M -concave. On 9725 products ($m \leq 4$, $H \leq 12$, supports of 2 to 91 points) ν was M -concave in every case, with no failure — so under the stronger reading the proof would become conditional on a measured-but-unproved lemma rather than simply broken. This is recorded as a measurement, not promoted to a step: M -concavity of the product is strictly stronger than (Q) and nothing here proves it. It is the natural next target, and if proved it removes the only unverifiable dependency in this paper.

What the controls can and cannot do

(Q) has no known counterexample — 80,471 instances before this session, zero failures — so no instance test on (Q) can go red, and every instrument available grades “is it true”, not “is it derivable from what I hold”. The controls below are therefore aimed at the *certifying chain* and at the *cited statement*, not at (Q) . Every block reports the number of objects enumerated and the size of the objects, not only the number of failures.

Instrument validation (known-present values first).

- The lean-verified criterion `13-det-reduction` (3×3 symmetric with nonnegative entries has at most one positive eigenvalue iff $e_2 \leq 0$ and $\det \geq 0$) was checked against an independent exact Sturm count of positive roots of the characteristic polynomial on 600 random nonnegative symmetric 3×3 matrices: 0 disagreements.
- The witness that killed `rlc-implies-13` was recovered independently by search: $M = \begin{pmatrix} 0 & 0 & 2 \\ 0 & 4 & 0 \\ 2 & 0 & 0 \end{pmatrix}$ has all three 2×2 principal minors ≤ 0 and $\det = -16$; the criterion rejects it and the exact count returns 2 positive eigenvalues, matching the recorded value -16 .
- `is_Mconvex` accepts the box slice $\{|\alpha| = 4, \alpha_3 \leq 2\}$ (10 points) and rejects $\{2e_1, 2e_2, 2e_3\}$ and that set plus $(1, 1, 0)$.
- The lattice-point enumerator for $k(a)$ agrees with a direct expansion of $\sum_w \prod_i [w_i]_z$ on 3177/3177 instances ($m \leq 3$).

- `prop:m3`'s recorded value is reproduced: $l = h = (2, 2, 2)$, $D = 6$ gives $k = (1, 3, 6, 7, 6, 3, 1)$, which is PF_2 and *not* concave ($1 + 6 > 2 \cdot 3$) — the recorded reason the $m = 2$ mechanism stops at $m = 3$.
- Proposition 11 was cross-checked against an independent graded enumeration that never forms the product: $\prod_i \tilde{P}_i$ equals the direct enumeration on all instances tested, and $k(a)$ equals the extracted coefficient on all instances tested (counts in Table 1).

An instrument defect, found by a false alarm against my own lemma. Lemma 18 is my own derivation, so it was given an instrument that does not share it: a random sweep over symmetric integer matrices satisfying its hypotheses. The sweep returned *one violation in 64*. The violation was not in the lemma. The matrix was $M = 4I_2$, whose eigenvalue 4 has multiplicity 2, and my “exact positive eigenvalue count” — built on `sympy`'s `Poly.count_roots`, which counts *distinct* real roots in an interval — had returned 1. So the instrument declared M admissible, and the lemma's conclusion correctly failed on it.

This matters more than a one-line bug, because that same function was the *independent validator* of the lean-verified criterion `l3-det-reduction`, i.e. it sat upstream of the licence for every Lorentzianity test in this paper. Its blind spot was exactly the repeated-eigenvalue case, which random integer matrices almost never produce — so the original 600-matrix validation returned 0 disagreements by *sampling around* the defect rather than by being right. Replaced with `Poly.real_roots()`, which returns real roots with multiplicity (and a real symmetric matrix has n of them), and the validation redone, now including an exhaustive sweep and a separate count of the repeated-eigenvalue matrices it had been blind to; see Table 1. The tell was a single violation of a lemma I had already proved: when a proof and an instrument disagree once, the instrument is the cheaper suspect, and here it was the guilty one.

The validator's own refusal test. `trustcheck` returns exit 0 on the registry for this session. That is only informative if it returns nonzero when something is wrong, so three violations were planted — a premise of a `proved` node downgraded to `computed`, a node pointed at a nonexistent file, a child's `role` field stripped — and it exited 1 on all three. Exit codes were read from the process, not from `$?` after a pipe.

C4, the control this session was for. The one citation that cannot be reopened is `CorollaryConvolution`. Rather than test (Q) again, test the recorded *statement*: search for f, g homogeneous in three variables with nonnegative coefficients such that $N(f), N(g)$ are Lorentzian but $N(fg)$ is not. Lorentzianity is decided exactly, in integer arithmetic, by Lemma 14 together with `l3-det-reduction` and a brute-force M -convexity check. Result:

779,641 pairs tested, 0 counterexamples.

The test is not vacuous as a test of *this* corollary: in 211,598 of those pairs at least one factor has ν_f *not* M -concave, so the pair lies outside the reach of `normalizedcoefficients` and the conclusion genuinely requires the convolution statement. Of the harvested factors, 129/879 in degree 2 and 84/929 in degree 3 have $N(f)$ Lorentzian while ν_f is not M -concave, which is also an independent confirmation of the recorded parenthetical that `normalizedcoefficients` is sufficient and *not* necessary.

This does not establish a citation right. It establishes that the recorded statement is not false in the regime the proof uses it, which is the failure mode that would kill Theorem 19.

C1–C3, refusal controls on the chain.

Control 22 (deleting an interior support point must be refused). Delete one interior point of $\text{supp } \tilde{P}^{(l,h)}$ and re-run the certification: 252 interior deletions, 252 refusals, every refusal reported as “support not M -convex”, so Step A’s hypothesis is what bites.

The first version of this control was not a control, and the repair is recorded. I first deleted *every* support point rather than every interior one, and 72 of 534 deletions did not refuse — which I was briefly ready to read as a hole in the chain. They are the boundary points: a box slice minus an extreme point can still be a box slice (an interval minus an endpoint is an interval), so those deletions are positive instances of Lemma 7, not controls. Over all 819 deletions, refusal occurred *exactly* when M -convexity was destroyed (534/534 agreement on the overlap tested). A second repair inside the repair: my own interior predicate scored a point with a zero coordinate as interior, because the exchange move out of that coordinate is unavailable and I had coded “unavailable” as “satisfied”. A point with a zero coordinate is on the boundary. With that fixed, 252/252.

Control 23 (non-laminar constraint family: the chain must fail to certify, and the conclusion must actually be false there). Replace the box family by the crossing family that adds $c \leq e_i + x_{i+1} \leq c'$ (so that $\{x_i, e_i\}$ and $\{e_i, x_{i+1}\}$ overlap without nesting), form the same three-variable graded homogenisation G , and run Step A–C on it. The chain must refuse. It is not enough that it refuses: the refusal must be *load-bearing*, i.e. there must be instances of the crossing family where log-concavity of the count is genuinely *false*, for otherwise the chain would be refusing a true statement and the control would carry no information. Both halves are reported in Table 1, with an explicit witness sequence that is not PF₂.

Control 24 (all-one coefficients on a non- M -convex support). `normalizedcoefficients` must not apply when the support fails to be M -convex even though all coefficients are 1. The sharp form of this control isolates the hypothesis: find supports with all coefficients 1 that are *not* M -convex and on which *every* Hessian condition nevertheless passes, so that the M -convexity hypothesis is the only thing standing between `normalizedcoefficients` and a false certificate. Four such supports were found in degree 3, the simplest being $\{X^3, W^3\}$: its support $\{(3, 0, 0), (0, 0, 3)\}$ admits no exchange, while every Hessian $[c_{\beta+e_i+e_j}]$ has a single nonzero entry and so at most one positive eigenvalue. The certification is refused, by the support test alone.

Recorded, because it is the same shape as the 1004 c2 “control that was no control”: of the five supports I first wrote down as non- M -convex, the instrument found that one, $\{(2, 1, 0), (0, 1, 2), (1, 1, 1)\}$, *is* M -convex and its N *is* Lorentzian. That row was a positive instance of the theorem, not a control, and I had labelled it by eye.

Regression gate. The 2026-10-04 c2 Lorentzianity sweep is re-run once, as a gate and not as this session’s evidence: $N(\tilde{P}^{(l,h)})$ Lorentzian on all 66 pairs $0 \leq l \leq h \leq 10$; the $m = 2$ and $m = 3$ products as recorded. Counts in Table 1.

A hedge that was measured, and what it would buy. If $\nu(\alpha) = \log c_\alpha$ were M -concave for the *product* $f = \prod_i \tilde{P}_i$, then `normalizedcoefficients` alone would give Proposition 13 and `CorollaryConvolution` could be dropped entirely, removing the only unverifiable dependency in this paper. Measured: M -concave on 9725/9725 products, $m \leq 4$, $H \leq 12$, supports of 2 to 91 points, no failure. The support half of that is a theorem — $\text{supp } \prod_i \tilde{P}_i = \sum_i \text{supp } \tilde{P}_i$ is a Minkowski sum of M -convex sets — but the inequality half is not proved here, and M -concavity

of the product is strictly stronger than (Q) . Recorded as a measurement and as the next target, not as a step.

8 Scope, and the fallback that was not needed

The brief for this session named a fallback — proving the converse of `thm:R`, that a slice’s width-vector set is M -convex only if the slice has a single half-width — to be taken up if step (C) had not closed in the first half of the session. Step (C) closed (Lemma 14 is unconditional, and it closed first, not last: the honest account is that step (C) was never the hard step, because m does not appear in it). No target was substituted and the fallback was not entered; `rem:converse` in `proofs/2026-10-04-width-vector-M-convexity.tex` stands open exactly as recorded.

The scope of Theorem 19 is (Q) itself, for all m , with the one dependency of §7. Its consequence for the programme is Corollary 20 and no more.

check	enumerated	result	object s
<i>Instrument validation — known-present values first</i>			
13-det-reduction vs. corrected exact eigenvalue count, random	600	0 disagreements	nonneg.
same, exhaustive over entries 0..4	15625	0 disagreements	
of those, matrices with a repeated eigenvalue	169	0 disagreements	the blind
instrument defect found and fixed: count_roots counts <i>distinct</i> roots, so $4I_2$ was scored as having one positive eigenvalue	1	replaced by real_roots()	see §7
rlc-implies-13 witness recovered independently by search	1	$\det = -16$, 2 pos. eigenvalues	as recor
M -convexity checker, known present / known absent	3	3 correct	3–10 po
$k(a)$ enumerator vs. direct expansion of $\sum_w \prod_i [w_i]_z$	3177	3177 agreements	$m \leq 3$
prop:m3 value $k = (1, 3, 6, 7, 6, 3, 1)$: PF_2 but not concave	1	reproduced	$D = 6$
Prop. 11: $\prod_i \tilde{P}_i$ vs. independent graded enumeration	1110	1110 agreements	$m \leq 3$
Prop. 11: $k(a)$ vs. the extracted coefficient	3439	3439 agreements	$D = 0..9$
trustcheck refusal test (planted boundary violations)	3	exit 1 on all 3; exit 0 clean	read from cess
<i>The two lemmas that are my own derivation, checked by instruments that do not share it</i>			
Lemma 14 symbolically, symbolic coefficients	45	45 agreements	$n \leq 4$, d
Lemma 18, random matrices satisfying the hypotheses	70	0 violations	$n = 2..5$
Lemma 18, exhaustive 3×3 , entries 0..3, both signs	1109	0 violations	
<i>C4 — refusal control on the cited statement, not on (Q)</i>			
pairs f, g with $N(f), N(g)$ Lorentzian: is $N(fg)$ Lorentzian?	779641	0 counterexamples	degrees
of which at least one factor has ν <i>not</i> M -concave	211598	so not vacuous	
harvested factors: $N(f)$ Lorentzian but ν_f not M -concave	129/879, 84/929	normalizedcoefficients not necessary	degrees
$N(fg) = N(f)N(g)$ on disjoint variables / on shared variables	1 / 1	true / false	54 / 20
<i>C1 — delete a support point of a bead</i>			
interior deletions, chain must refuse	252	252 refusals , all “support not M -convex”	support
all deletions, interior and boundary	819	728 refusals	
refusal occurred <i>exactly</i> when M -convexity was destroyed	534	534 agreements	
<i>C2 — non-laminar crossing family: the chain must fail to certify</i>			
crossing families with support <i>not</i> M -convex	1980	1124	$m = 2, 3$
crossing families with $N(G)$ <i>not</i> certified Lorentzian	1980	1557 refused	
crossing instances where log-concavity is genuinely false , with full interval support	43242	7200 failures	$m = 2, 3$
strongest witness $k = (7, 12, 15, 19, 15, 12, 7)$: $15^2 = 225 < 228 = 12 \cdot 19$	1	the refusal is load-bearing	$m = 3$, $n = 7$
<i>C3 — all-one coefficients on a non-M-convex support</i>			
supports, all coefficients 1, not M -convex, <i>every</i> Hessian condition passes	4	all refused, by the support test alone	degree 3
my own candidate list, audited by the instrument	5	1 was M -convex — not a control	degrees
	14		
<i>Regression gate (re-run once; not this session’s evidence)</i>			
$N(\tilde{P}^{(l,h)})$ Lorentzian, all $0 \leq l \leq h \leq 10$	66	0 failures	support
$N(\tilde{P}_1 \tilde{P}_2)$, $h_i \leq 5$ / $N(\tilde{P}_1 \tilde{P}_2 \tilde{P}_3)$, $h_i \leq 3$	441 / 1000	0 failures	support